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Editor-in-Chief
Journal of Mechanical Science and Technology (JMST)
Korean Society of Mechanical Engineers

Dear Editor-in-Chief,

I am pleased to submit the enclosed manuscript, entitled

"Method-Induced Uncertainty in Gear Bending Fatigue: Factorial Decomposition of Size-and-Surface Factor Sensitivity Under a Common ISO Stress Baseline",

for consideration for publication as an original research article in the *Journal of Mechanical Science and Technology*.

The manuscript quantifies how much of the scatter in gear bending-fatigue life estimates is inherited from the analyst's methodological choices rather than from the material itself. A 144-combination full-factorial design spanning five methodological switches—S–N curve source, mean-stress correction, cumulative damage rule, size-and-surface standard, and surface roughness—is decomposed with a censored-likelihood accelerated failure-time (AFT) model. Because 84 of the 144 combinations are run-outs (58% of the dataset), ordinary ANOVA/Sobol' approaches, which must discard censored entries, systematically distort the ranking; the AFT framework retains all observations through the EM algorithm. The decomposition attributes the method-induced variance to six contributions—55.3%, 22.6%, 18.2%, 2.5%, 1.2%, and 0.3%—with bootstrap resampling (2000 resamples, seed 42) confirming the stability of the ranking. A Shapiro–Wilk test ($W = 0.962$, $p = 0.056$) supports the residual-normality assumption, and a model-selection comparison (ΔAIC $= +77.1$) confirms the log-normal AFT specification over the Weibull alternative. The paper reports 11 figures and 6 tables.

We believe these results are directly relevant to JMST's readership in mechanical design and fatigue assessment: they show that a large fraction of reported fatigue-life scatter can be traced to factor choice and calibration, and they provide a reproducible variance-decomposition workflow (Python scripts, seed-fixed randomness, Monte Carlo validation, and a complete reproducibility package) that any fatigue-engineering laboratory can apply to its own gear geometries, materials, and loading conditions.

This manuscript is original, has not been published previously, and is not under consideration by any other journal. The author declares no competing interests.

Generative AI tools assisted with LaTeX formatting, Python code implementation, and English language editing.

Thank you for your consideration. I look forward to the reviewers' feedback.

Sincerely,

Zhilei Chen
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